

Harpoon anti-ship missiles.

The Indian Navy's attempts to acquire this capability were realised in the mid-'90s when its Soviet-built Kilo class submarines (acquired between 1986 and 2000) were sent to Russian shipyards for mid-life refits. It decided to equip them with the submarine-launched variant of the 220-km range Klub anti-ship cruise missile and India became the first customer for the yet untested missile. The missile, designated SS-N-27 'Sizzler' by NATO, had already been ordered in 1998 for three Russian-built Talwar class frigates.

While the ship-launched missile was inducted without problems, the navy has been frantically trying to rectify the defect in the submarine-launched variant over the past three years without success. The Klub missiles were successfully test-fired in the Baltic Sea in 2001 and 2002. The problems surfaced when the missiles were test fired in Indian waters: minutes before reaching its target, the missile wobbled before diving into the water.

A detailed analysis of the missile recently carried out by Russian experts revealed that the problem was in the Kilo class submarine.

Every mobile military platform

has, what is called, a gyroscope—a device which spins at high speed and tells a platform its roll, pitch and yaw—basically where it is heading, how its motion is changing and its axis of stabilisation. A gyroscope which stabilises the missile in flight is also the heart of a missile's on-board guidance system. Since the gyro cannot be kept switched on all the time, the coordinates are pre-fed to the missile from the gyro during pre-launch checks. When the Klub missile breaches the water surface, its gyro tells the missile its position and points itself in the direction of the target. However, a critical time-lag in the interface between the submarine's gyro and the missile, was pre-fed the wrong coordinates.

So, when the missile flew over the

sea surface, the guidance coordinates fooled it into believing it was actually flying at an incline. So the missile's onboard gyroscope stabilised itself. This battle between the onboard coordinates and gyroscope continued several times and the manoeuvres exhausted the missile battery leading to a complete power loss.

New replacement gyros for the navy's submarines are now among the top of the list of items India desperately needs from Russia and the issue has already been raised at several high-level meetings. The current problems with the firing platform, however, have placed under a cloud on the efficacy of more purchases: the navy has already ordered more variants of the missiles. In 2006, SIPRI says, India ordered 28 land-attack versions of the Klub missiles for Rs 844 crore (\$182 million).

The Indo-Russian BrahMos corporation is retrofitting one Kilo class submarine with the indigenously built 300-km range supersonic cruise missile. Yet the retrofit is a complicated procedure and is, therefore, unlikely to be repeated on the other submarines. Till then, the navy says it has little option but to swim or sink with the Klub. ■

READING IT WRONG

■ **THE PROBLEM:** Missile fired from the submarine ditches into water before hitting target.

■ **THE REASON:** Delayed transmission of data between submarine and missile before launch

■ **THE SOLUTION:** New gyros for all submarines.

HOW IT FAILED

The submarine launched missile didn't hit its target

